

IEC 60870-5-104 driver

This module is a client (controlling station) driver for the IEC 60870-5-104 TCP telecontrol protocol, used in electricity transmission and distribution, connecting Ignition to an RTU or substation gateway.

Version 0.1.0 Ignition 8.1.33+ Runs on: any gateway that reaches the RTU network

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1. Overview

The driver is a standalone OPC UA device type; it does not require any other SparkBridge module. Control-direction commands (single/double/step/setpoint), counter interrogation, file transfer, parameter loading, and TLS (secure 104) are not implemented in v1.

2. What You Need

- Ignition 8.1.33 or newer. The signed release is built and tested against Ignition Module SDK 8.1.54.
- An RTU or substation gateway reachable over TCP, default port 2404.

3. Install

Install the signed IEC 60870-5-104 module through Config > Modules > Install or Upgrade a Module; no build step is required.

4. License

See the README in the bundle for licensing terms.

From release 2.7.0, every SparkBridge module runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). A verified key overrides the trial entirely and the trial clock is then ignored. Drop the key at `<data>/sparkbridge-license.key`; each module re-checks the file every 30 seconds, so installing, replacing, or removing a key takes effect with no restart.

5. Configure

Config > OPC UA > Device Connections > Create new Device > IEC 60870-5-104.

Setting	Meaning, default
Hostname/IP, Port	RTU or substation gateway; port default 2404.
Backup Hostname/IP, Backup Port	Second server of the redundancy group; blank for a single server.
Common Address	Common address of ASDU, default 1.
Originator Address	Echoed on responses; 0 means unused, default 0.
Window k / Window w	Max unacknowledged I-frames (default 12) and the acknowledgement threshold (default 8, must not exceed two thirds of k).
t1 / t2 / t3 (s)	Ack/confirmation timeout (15), S-frame ack delay (10), keepalive idle time (20).
Interrogate On Connect	Default true.
Clock Sync On Connect	Default false.
RTU Time Zone	Zone the RTU expresses CP56Time2a in, default UTC. Getting it wrong silently offsets every time tag.

Addressing: a bare IOA such as 100, or structured form 1.2.4 (high.middle.low). No type suffix; the point's type is decided by the RTU. There is no browse service for undiscovered points; browse shows only IOAs already reported.

6. Operate

With Interrogate On Connect true (default), the driver requests a general interrogation each time it connects so its point set is populated without waiting for spontaneous reports. Watch the RTU Time Zone setting whenever an RTU is relocated or its clock configuration changes; a mismatch silently offsets every timestamp rather than raising an error.

7. Verify

- Reproduce the bundled live-proof harness: `java -cp common/build/libs/common-0.1.0.jar com.greenguard.iec104.tools.Iec104Probe --host <ip> --port 2404 --ca 1 --seconds 70 --clocksync --t3 1000.`
- Check `GET /data/com.greenguard.drivers.iec104/status.json` for lane state and data age.

8. Troubleshoot

Symptom	Cause / fix
Timestamps are off by a fixed offset	RTU Time Zone does not match the zone the RTU expresses CP56Time2a in; correct it and reconnect.
Points never populate on connect	Confirm Interrogate On Connect is true, or issue a manual general interrogation.
Unexpected disconnects under load	Review Window k / Window w and the t1/t2/t3 timers against the RTU's own configuration; a mismatch on these values is a common source of link resets.

9. Reference

Module id	<code>com.greenguard.drivers.iec104</code>
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Default port	TCP 2404
Status endpoint	GET /data/com.greenguard.drivers.iec104/status.json
Not implemented in v1	Control-direction commands, counter interrogation, file transfer, parameter loading, secure 104 (TLS)

10. Support

Docs page section: </sparkbridge/docs#iec104>. Contact: </sparkbridge/contact>. Release downloads and checksums: [GitHub releases](#).